

BENJAMIN KUTER

bmkuter.github.io | Norwell, MA | bmkuter@gmail.com | (617)-840-4388

WORK EXPERIENCE

Firmware Engineer

P & E Microcomputer Systems Inc. | Watertown, MA | 2022 – Present

Responsible for low-level firmware drivers for flagship products; networking; cryptography and hardware security; linkers and memory space; production programming.

Research Technician

Priscilla Brastianos Lab, Massachusetts General Hospital | Boston, MA | 2017 – 2020

Genomic sequencing of metastatic brain cancer.

SKILLS

Python · TypeScript · React · Groq AI · OAuth2 · CSS3 · HTML5 · ESP-IDF · ESP32-S3 · ESP32-C6 · ESP Mesh Lite · ESP-NOW · FreeRTOS · Edge Impulse · NimBLE · SQLite · WiFi · Web Interface · Blockchain · TCP · I2C · UART · HTTPS · CMake · Sensors · Embedded C · Jenkins

PROJECTS

Production Programmer & Debugger

NXP i.MX RT1064 · RTOS · GPIO · Ethernet · USB · Cryptography · Linkers · Production Programming · Debugging · WiFi · ARM M7

Designed and implemented WiFi networking and core RTOS for the Cyclone FX Programmer and Debugger on NXP i.MX RT1064 ARM M7

- Designed and implemented WiFi networking for seamless communication with host PCs.
- Developed core RTOS for efficient management of system resources and tasks.
- Implemented board-level drivers for NXP i.MX RT1064 MCU, including RAM mapping, flash mapping, and GPIO control.
- Optimized firmware for improved performance and reduced power consumption.
- Developed and integrated production programming, validation, and debugging tools for the Cyclone FX Programmer and Debugger.
- In charge of debugging, stress testing, and edge-case testing.

ESP32-S3 Voice-Controlled Audio Recorder

ESP32-S3 · Edge Impulse · NimBLE · I2S · LittleFS · FreeRTOS · C · Python

Battery-powered voice recorder with on-device wake word detection and BLE control

- Designed and implemented a custom-trained neural network for on-device wake word detection
- Developed a low-power audio pipeline with automatic light sleep and deep sleep functionality
- Implemented a double-buffered SD card read and I2S DMA write pipeline for audio playback
- Integrated BLE remote control with a full command set for recording, playback, and file management

ESP32-S3 Camera

ESP-IDF · ESP32-S3 · OV3660 · WiFi · mDNS · Python · HTTP API · MJPEG · Web Interface

WiFi-enabled camera system for high-resolution image capture and live preview.

- Designed and implemented a WiFi-enabled camera system using ESP-IDF and OV3660 camera module.
- Developed a web-based interface for camera settings and live preview with MJPEG video stream.
- Created a Python client for image capture using mDNS hostname resolution and HTTP API endpoints.
- Optimized performance for ~1.5 second capture-to-transfer time and ~10 FPS MJPEG stream.

Weather Mesh Lite

ESP Mesh Lite · ESP-NOW · Blockchain · C++ · I2C · TCP · SHA-256 · NVS Storage

Distributed sensor network for collecting and aggregating weather data using blockchain-like records and ESP Mesh Lite technology.

- Designed and implemented a blockchain module for handling block creation, hashing, and history management.
- Developed a mesh networking module using ESP-NOW for inter-node communication and broadcast messages.
- Implemented a consensus and election module with leader election and Proof-of-Participation for block creation and sensor data validation.
- Integrated WiFi networking for remote TCP messaging and connectivity with external servers and interfaces.

EDUCATION

Master of Science in Electrical & Computer Engineering, Cybersecurity focus

Boston University, LEAP, 3.9 GPA | Boston, MA | 2020 – 2022

Bachelor of Arts, Neuroscience

Macalester College | St. Paul, MN | 2013 – 2017

AWARDS

Engineering Graduate Scholarship

Boston University

The Center for Engineering in Medicine (CEM) Internship (Hematology)

PROJECTS CONTINUED

Daily Idea Planner

JavaScript TypeScript React Next.js Groq AI Transformers.js localStorage Google Drive OAuth2 ES6+ CSS3 HTML5

A minimalist browser-based journal for brainstorming startup ideas with AI idea generation and interactive graph visualization.

- Designed a browser-based journal with infinite idea slots, automatic collapse, and favorite marking for brilliant ideas.
- Implemented localStorage persistence for data saving, a 30-day rolling history, and calendar navigation with visual indicators.
- Built a smooth initialization overlay with 7-step progress tracking and real-time status updates.
- Developed a performance-optimized AI integration with cached clusters, non-blocking auth, and auto-close functionality.
- Created a calendar panel with edit historical entries, large textarea editing, and resizable layout with saved settings.

GrowPod

ESP32-S3 ESP-IDF Python Adafruit SQLite I2C GPIO PWM UART HTTPS

Scalable Deep Water Culture System for at-home food production and scientific plant research

- Designed and implemented a low-power, water-recycling hydroponic system with ESP32-S3 and off-the-shelf peristaltic pumps
- Developed a Python GUI console with device discovery, multi-tab interface, and historical data viewer
- Improved energy efficiency through smart scheduling and real-time power monitoring with INA219
- Implemented complete cycle control with custom level sensing and flow monitoring for water recycling

Coffee Bean Smart Doser

ESP32-C6 CMake FreeRTOS I2C OLED Display Adafruit Seesaw Rotary Encoder NAU7802 24-bit ADC Strain Gauge Load Cells ESP-IDF

Precision coffee bean dosing system with strain gauge scale integration and OLED display.

- Designed and implemented a multi-threaded architecture using FreeRTOS tasks for responsive operation.
- Developed a centralized display refresh system with minimal flickering and targeted updates.
- Integrated strain gauge load cells for precision weight measurement and implemented real-time weight monitoring.
- Implemented scale taring functionality with visual confirmation via status overlay and immediate weight display update.

Folio Resume Builder

React 18 TypeScript Vite Tailwind CSS Zustand Groq API Vercel

A structured resume content system for composing tailored resumes without rewriting from scratch.

- Designed a three-layer model for resume composition, separating Vault, Folio, and Export layers.
- Implemented a heuristic section detection system for parsing CVs and importing structured content.
- Built a Vault repository for permanent storage of imported and manually created content items.
- Developed a Resume Editor with features like section arrangement, node reordering, and tag management.
- Created a Layout Canvas for free-canvas A4/Letter page layout and multi-page support.

PUBLICATIONS

- Prakadan SM, Alvarez-Breckenridge CA, Markson SC,... Kuter BM ,... Brastianos PK. Multicellular responses to PD-1 blockade within the tumor microenvironment of leptomeningeal metastases.
- Shih DJH, Nayyar N, Bihun I,... Kuter BM ,...Brastianos PK. Genomic characterization of human brain metastases identifies novel drivers of metastatic lung adenocarcinoma . Nature Genetics. 2020.
- Ippen FM, Grosch JK, Subramanian M, Kuter BM ,...Brastianos PK. Targeting the PI3K/Akt/mTOR-pathway with the pan-Akt inhibitor GDC-0068 in PIK3CA-mutant breast cancer brain metastases . Neuro Oncol. 2019 Jun 7. doi: 10.1093/neuonc/noz105.
- Ippen FM, Alvarez-Breckenridge CA, Kuter BM ,...Brastianos PK. The dual PI3K/mTOR-pathway inhibitor GDC-0084 achieves antitumor activity in PIK3CA -mutant breast cancer brain metastases . Clinical Cancer Research. 2019 Feb. doi: 10.1158/1078-0432.CCR-18-3049.
- Nayyar N, White MD, Gill CM,... Kuter BM ,...Brastianos PK. MYD88 L265P mutation and CDKN2A loss are early mutational events in primary central nervous system diffuse large B-cell lymphomas. Blood Adv. 2019 Feb. doi: 10.1182/bloodadvances.2018027672.

RESEARCH EXPERIENCE

Research Technician

Priscilla Brastianos Lab | Massachusetts General Hospital

Conducted & optimized single-cell sequencing of cerebrospinal fluid from metastatic brain cancer patients using a novel technique developed with collaborators at the Massachusetts Institute of Technology.

Undergraduate Capstone Thesis

Macalester College | St. Paul, MN

Developed & conducted a self-run study on the effects of electric fields on neuronal properties in crayfish that was later presented at a national conference (Society for Neuroscience 2016).

The Center for Engineering in Medicine Internship

Biju Parekkadan Lab | Shriners Children's Boston

Prototyped devices to treat patient-derived blood with stem cell growth factors.